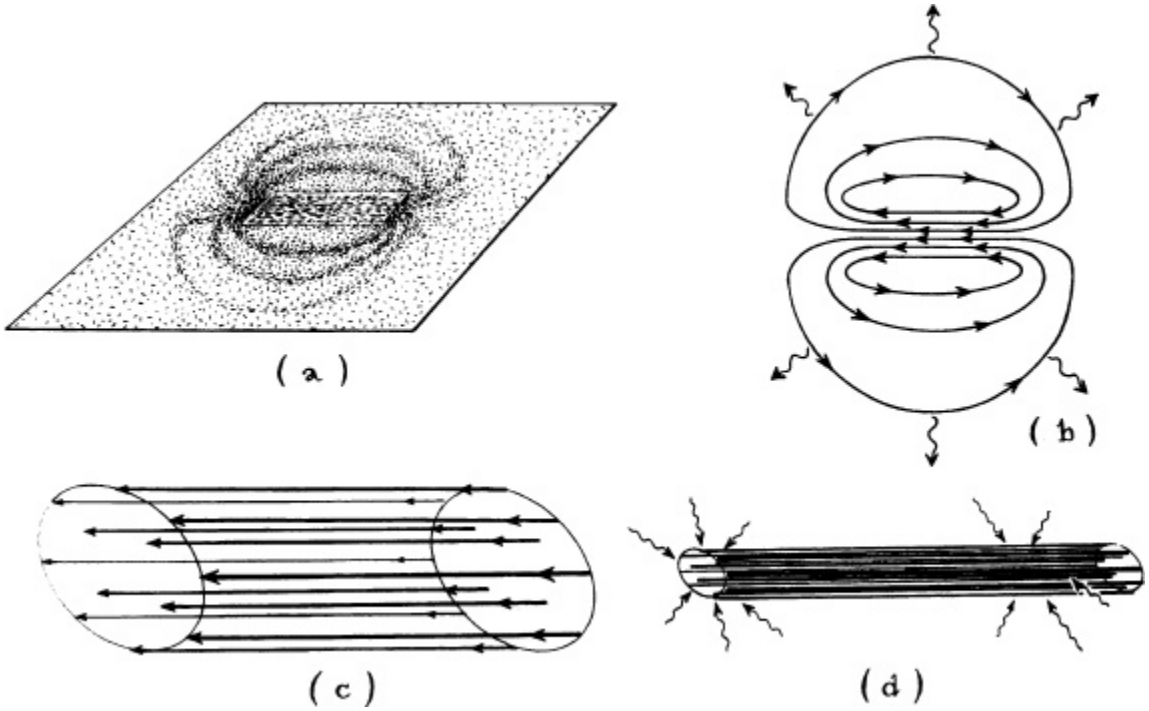
their repulsion will make the field lines explode (Figure 7.tb). All this was familiar to me from my undergraduate studies. Wheeler reminded me of it in a long, private discussion in his Princeton office. He then described a recent discovery by his friend Professor Mael Melvin at Florida State University in Tallahassee.



7.1 (a) The magnetic field lines around a bar magnet, made visible by iron filings on a piece of paper with the magnet below it. (b) The same field lines, with the paper and the magnet removed. Pressure between adjacent field lines makes them explode in the directions of the wavy arrows. (c) An infinitely long, cylindrical bundle of magnetic field lines whose field is so intense that its energy creates enough spacetime curvature (gravity) to hold the bundle together, despite the repulsion between field lines. (d) Wheeler's conjecture that when the bundle of field lines in (c) is squeezed slightly, its gravity would become so strong as to compress the bundle into implosion (wiggly lines).

Melvin had shown, using Einstein's field equation, that not only can magnetic field lines be held together against explosion by the iron in a bar magnet, they can also be held together by gravity without the aid of any magnet. The reason is simple: The magnetic field has energy, and its energy gravitates. [To see why the energy gravitates, recall that energy and mass are "equivalent" (Box 5.2): It is possible to convert mass of any sort (uranium, hydrogen, or whatever) into energy; and conversely, it is possible to convert energy of any sort (magnetic energy, explosive energy, or whatever) into mass. Thus, in a deep sense, mass and energy are merely